

Effects of Household and Individual Factors on Travel Mode Choice

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Executive Summary

This study investigates how household demographics (income level, household size, region and car ownership) and individual characteristics (age groups and employment status) influence the choice of travel mode. The analysis was based on the data from the Victorian Integrated Survey of Travel and Activity (VISTA), including the Household, Person, Journey to Work and Journey to Education datasets.

Comprehensive data preprocessing was conducted, including data integration, feature engineering and outlier detection to ensure data integrity and consistency. Normalised mutual information (NMI) was then computed to assess correlation between the selected features and travel modes, serving as a foundation for further supervised learning phases. Two classifications models, k-Nearest Neighbours (KNN) and Decision Tree (DT) were implemented to predict individuals' travel mode choices.

The results indicated that car ownership is the most influential factor affecting travel mode choice. Households possessing one or more vehicles, with at least one member holds a driving licence, predominantly rely on private transport modes. In contrasts, the dataset provided limited evidence explaining the determinants of public or active travel mode choices, largely due to insufficient sample size. Further research with a larger and more balanced dataset is recommended to allow a more comprehensive understanding of these travel behaviours.

Introduction

This report investigates how household and demographic characteristics influence people's choice of travel mode in Victoria. The central question guiding the analysis is: Which household and individual factors, such as income, car ownership, household size, age, and employment status, most strongly affect how people travel to work or education. Understanding these relationships is crucial for developing more efficient and equitable transport systems that meet the diverse needs of the population.

The study draws on data from VISTA (State of Victoria, 2025). This dataset provides comprehensive information on household composition, demographics, trip details, and travel modes across the state. Its detailed structure allows for quantitative analysis using data-driven and machine learning techniques to identify significant predictors of travel behaviour.

The purpose of this analysis is to generate insights that can inform evidence-based urban and transport planning. By identifying which groups are more reliant on private vehicles and which factors encourage public or active transport, policymakers can design targeted interventions to reduce congestion, improve accessibility, and support sustainability. In doing so, this research contributes to the broader goal of promoting efficient, inclusive, and environmentally responsible mobility in Victoria.

Methodology

Data Pre-Processing

1. Data Integration

Multiple datasets from VISTA – including household, person, journey_to_work and journey_to_education were merged using common household and person identifiers (i.e., hhid, persid). This integration allowed linking household demographics, individual characteristics and trip-level information, enabling multidimensional analysis of choice of travel mode.

2. Feature Engineering

Features were created with the research aim of investigating how household demographics and individual characteristics influence travel mode choice.

1. **Car Ownership:** Car ownership was determined by combining vehicle availability within household and possession of driver licence to accurately represent household-level access to private transport. A household was identified as owning a car if reported to own one or more vehicles ($\text{totalvehs} \geq 1$) and at least one member held a valid driver licence (only full licence and green probationary licence). For people that hold a valid licence, they are considered as potential drivers if they also possess one or more vehicles. And if a household has at least one potential drivers, then all members would be potential passengers. The two datasets were then merged using the household identifier (hhid) to determine whether each household met both ownership and licence conditions. Missing or unmatched records were treated as no ownership cases to avoid overestimating private vehicle access. This approach provided a more realistic measure of household car availability by ensuring that physical vehicle ownership was paired with the ability to drive.
2. **Bicycle Ownership:** Bicycle ownership was derived from household-level counts of bicycles (totalbikes) relative to household size (hysize) to reflect accessibility within the household. Both variables were converted to numeric format to ensure consistent comparison. Households were categorised into three groups: 'No bikes' ($\text{totalbikes} = 0$), 'Insufficient bikes' ($0 < \text{totalbikes} < \text{hysize}$), and 'Sufficient bikes' ($\text{totalbikes} \geq \text{hysize}$). This classification captured not just the presence of bicycles but also their adequacy relative to household members, providing a more meaningful indicator of cycling accessibility for travel behaviour analysis.

3. Feature Discretisation

Each feature was cleaned and categorised to use for further analysis.

1. **Income Level:** The household income values were first converted to string format and trimmed of extra spaces to avoid mismatched entries caused by inconsistent text formatting. A set of predefined tokens (e.g. "", "NA", "N/A", etc.) was used to identify missing or invalid entries which were replaced with NaN to accurately flag missing data. To assess whether missing income values represented genuine non-income households or data quality issues, the household and person datasets were merged using the common 'hhid'. Each household was then evaluated based on the income and employment characteristics of its members. Specifically, personal income (persinc) and employment status (emptype) were used to check whether all members had no income and were not in the labour force or unemployed. This diagnostic process helped distinguish between truly income-less households and those with incomplete reporting.
2. **Employment Status:** Employment status was categorised into Full-time, Part-time, Not in labour force/Retired, Unemployed and Student since there were too many raw employment labels (e.g., "Not Yet at School", "Full-time TAFE", "Retired") overlap conceptually. Merging them avoids redundancy and ensures each group represents a distinct economic participation level.

3. **Household Size:** Household size was binned into four different categories: (1, 2, 3-4, 5+) to capture household size effects while ensuring sufficient sample size per bin. Instead of treating every household size as a separate category (which could make the data too fragmented), the values were grouped to reflect typical living arrangements (single-person, couple, medium-sized family, and large family). This helps reduce noise and ensures there are enough observations in each group for meaningful comparison.
4. **Region:** This variable was derived from 'homesubregion_ASGS', grouped into Inner, Middle, Outer Melbourne to explore how different geographical location influenced travel mode choice through differences in transport accessibility and urban density.
5. **Age Group:** Age groups were converted into decade bins (0-9, 10-19, ..., 70+) from 'agegroup' column to reduce noise and capture life-stage patterns in travel behaviour while ensuring sufficient observations per category.
6. **Travel mode:** Main mode of travel was re-classified into three main categories: Private (vehicle driver/passenger, taxi, rideshare), Public (bus, train, tram) and Active (walking, cycling, running, scooter) to simplify the analysis since there were too many specific choices of travel in the raw data which may affect the model performance. Thus, the re-classification reduced complexity of the raw outcome while reflecting the key behavioural differences in how individuals travel.
7. **Distance:** To categorise trip distances for comparative analysis, journey distances were divided into six discrete intervals: 0–50 km, 50–100 km, 100–150 km, 150–200 km, 200–250 km, and 250 km or more. This binning approach simplified the continuous distance variable by grouping similar journey lengths together, making it easier to identify and compare travel behaviour patterns across different distance ranges.

3. Outlier Detection and Handling

1. **Distance:** Outlier detection for journey distance was performed to ensure that distance bins accurately represent realistic travel behaviour. Records with non-positive values (≤ 0 km) were identified using a simple conditional filter, as negative travel distances are physically impossible and cannot occur in real travel scenarios. This approach guaranteed that subsequent analyses of distance patterns and mode choice were based on realistic travel measurements while ensuring that subsequent binning reflect genuine travel patterns.
2. **Household Size:** For household size, the interquartile range (IQR) method was used to identify values that deviated significantly from the majority of observations. A box plot was also used to visually confirm the presence of potential outliers and assess their influence on the overall data spread. Records lying outside the upper or lower IQR bound were flagged for review.
3. **Group & Income Level:** Outlier detection for income level and age group focused on identifying inconsistent or invalid values, as both features are ordinal rather than continuous. To ensure data integrity, all unique values in each variable were printed and manually reviewed to detect irregular entries such as blank fields or non-standard formats. For **income level**, this process helped confirming that all categories followed consistent range formats (eg, "\$0-\$9999") and excluded unexpected values such as "Unknown". For **age group**, unique category labels were inspected to verify that all values aligned with valid five-year intervals and there is no unreasonable data such as negative age or two hundred years old. This inspection ensured both variables were clean, interpretable and suitable for later binning, correlation analysis and machine learning models.

Correlation

To explore and evaluate how household and individual characteristics are associated with travel mode choice, the **normalised mutual information (NMI)** metric was applied. NMI measures the correlation and shared information between two variables on a scale from 0 (no relationship) and 1 (perfect association). This method is suitable for categorical data and can detect both linear and non-linear associations.

1. Applied method

Each predictor, including income level, employment status, household size, age group, region, car ownership and distance group was evaluated against target variable (travel mode group) using `normalized_mutual_info_score` function from `scikit-learn` library. The NMI values were normalised by the smaller entropy value between the predictor the target, ensuring comparability across variables with different category counts. This approach provided a consistent, dimensionless measure of statistical dependence for all features. The use of NMI was chosen since it is suitable with categorical predictors without requiring numerical encoding, captures both non-linear relationship and non-monotonic relationships which simpler correlation measurement such as Pearson may fail.

2. Alternative methods

Pearson's correlation was eliminated as visual investigation of scatter plots between each predictor (e.g. income level, age group, region and employment status) and travel revealed no clear linear trends. For example, as shown in figure 1, age group and employment status showed no proportional change in travel mode choice. Similar non-linear and categorical patterns were observed across other predictors.

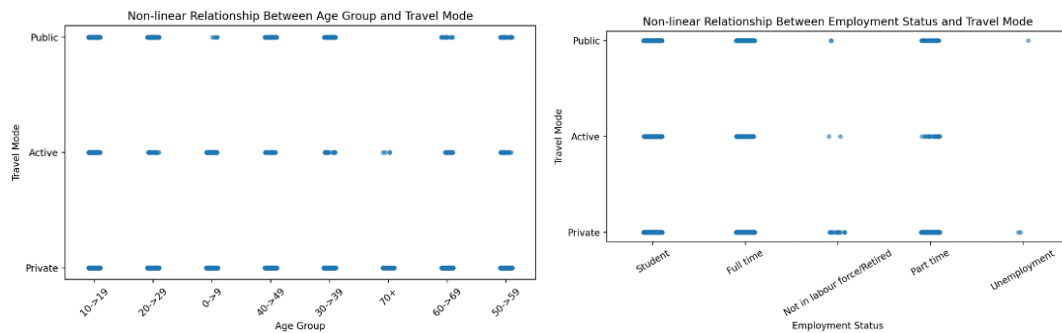


Figure 1: Scatter plot between travel mode and age group (left) employment status (right)

Cramer's V and **Chi-square** tests were also evaluated for categorical association but were not used since their values depend on contingency table size, making them less comparable across multiple predictors.

Supervised Machine Learning

All explanatory variables in the pre-processed dataset were first encoded into numerical values using one-hot-encoding scheme to maintain compatibility with the models. The dataset was then randomly partitioned into training and testing sets with 80:20 ratio. Each selected model was trained on the training set and subsequently evaluated on the testing set to report model performance via confusion matrix, accuracy, precision, recall and F1 score. To optimise model performance and minimise the risk of overfitting, k-fold validation was applied to tune hyperparameters, with five folds by default. The results were generated using the optimal hyperparameter identified in the validation process. The models examined were as follows:

1. Applied methods

Since the response variable is categorical, a classification approach was employed in the supervised machine learning process. The **K-Nearest Neighbours (KNN)** model was trained using the training set. Utilising k-fold validation, the hyperparameter k was evaluated by the model average accuracies tested on different values of k. The value that yielded the highest accuracy would be selected as the optimal hyperparameter. Subsequently, the KNN model was refitted on the training set using the optimal value.

Decision tree is another widely used supervised machine learning technique. Same as KNN model, the model was first trained using the hold-out set. The hyperparameter `max_depth` was selected to be tuned for enhancing model performance. Other commonly used hyperparameters include `min_samples_split` and `criterion`. One advantage of decision trees is that the decision made at each stage can be explicitly shown. This visualisation can illustrate the importance of each feature in the classification process.

2. Alternative method

Regression represents another main approach in supervised machine learning was considered as an alternative. However, linear regression is only applicable to continuous response variable, which was not suitable for the selected outcome. As a result, linear regression was deemed inappropriate and was therefore excluded from further analysis in this research.

Results Exploration and Analysis

Data Pre-processing

1. **Missing Income Level:** An assessment of missing household income values revealed clear and interpretable patterns rather than random data loss. Initially, the household and person datasets were merged and examined together. This analysis showed that most households with missing income consisted entirely of members reporting “Nil income” or “Negative income” and they were also not in the labour force or unemployed (Figure 2 left).

To further verify this observation, a diagnosis was conducted to explore the reason of these missing data (Figure 2 right). Out of all households with missing income data, the majority (73 households) consisted entirely of members with no personal income and not participating in the labour force (e.g., retired persons, students, or home duties). A smaller portion (11 households) had at least one working member, while only 5 households contained a mix of earners and non-earners.

Based on this finding, missing income values were imputed as zero income to accurately capture the economic status of these households while maintaining dataset completeness. Alternative imputation methods such as filling missing values using the mean or median income, or imputing by demographic groups (e.g., age or employment), were not used since the missing household income values reflect a systematic pattern tied to employment and income status, not missing at random. Thus, alternative methods would introduce bias and misrepresent true non-earning households. The chosen approach preserved both the integrity and interpretability of the income variable for subsequent analysis.

	persid	hhid	persinc	hhinc_group
64	Y24H5740504P01	Y24H5740504	Nil income	NaN
65	Y24H5740504P02	Y24H5740504	Nil income	NaN
303	Y24H5760304P01	Y24H5760304	Nil income	NaN
308	Y24H5760310P01	Y24H5760310	Negative income	NaN
309	Y24H5760310P02	Y24H5760310	Nil income	NaN
...	...	...	...	...
7780	Y24H6220422P02	Y24H6220422	Negative income	NaN
8000	Y24H6240211P01	Y24H6240211	Nil income	NaN
8001	Y24H6240211P02	Y24H6240211	Nil income	NaN
8017	Y24H6240236P01	Y24H6240236	Nil income	NaN
8152	Y24H5820333P04	Y24H5820333	Nil income	NaN

134 rows x 4 columns

Summary of Missing Household Income Reasons		
	Reason	Number of households
0	All members have no income & not working	73
1	At least one member is working	11
2	Some members no income & not working	5

Figure 2: Merged dataset showing income of each member with missing household income (left), and table summary showing number of households with reasons of missing household income (right)

2. **Distance Outliers:** An examination of the journey_distance variable revealed nine records containing negative values, which are logically invalid as travel distances cannot be below zero. These errors were likely caused by data entry or measurement issues. The identified records were removed from the dataset to ensure data accuracy and prevent distortion in subsequent analyses. All remaining records were retained, including those representing long-distance trips, as these values reflect plausible long commutes rather than true outliers. Keeping these extreme but valid observations helps preserve the real variation in travel behaviour across regions, particularly for households located in outer or regional areas such as Greater Melbourne. This cleaning step ensured that the distance data accurately represented feasible travel patterns while maintaining the diversity of trip lengths necessary for meaningful comparison across travel modes.

3. **Household Size Outliers:** The box plot analysis of household size revealed two outlier observations with unusually large family sizes of 8 and 10 members (Figure 3). While these values fall outside the interquartile range, they were retained in the dataset because they represent realistic cases rather than data entry errors. Large households, although less common, are still valid within the population and may reflect specific socio-demographic contexts such as multi-generational living arrangements. Keeping these valid extreme values ensures that the analysis continues to capture the full range of household structures, which is essential when investigating how family size influences travel behaviour and mode choice.

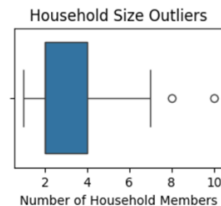


Figure 3: Box plot illustrating distribution of household size

4. **Age group & Income Level Outliers:** A review of the unique values in both income level and age group variables revealed that all entries followed the expected categorical structure and fell within valid ranges (Figure 4). No anomalous or invalid values were detected, indicating consistent data formatting and appropriate coding across the survey responses.

For income level, all categories aligned with realistic income brackets and contained no irrelevant text entries or missing labels. Similarly, the age group distribution showed a continuous and logical progression from early childhood (0–4) to older adulthood (100+), with no gaps or irregular groupings. Since the data exhibited no inconsistencies or absurd values, all observations were retained. This ensured that the full demographic and socioeconomic variation within the dataset was preserved for subsequent correlation and predictive analyses.

```
Unique values for Household Income Level:
['$1,250-$1,499 ($65,000-$77,999)' '$500-$649 ($26,000-$33,799)'
'$1-$149 ($1-$7,799)' '$2,000-$2,499 ($104,000-$129,999)'
'$1,750-$1,999 ($91,000-$103,999)' '$4,000-$4,499 ($208,000-$233,999)'
'$1,500-$1,749 ($78,000-$90,999)' '$3,500-$3,999 ($182,000-$207,999)'
'$2,500-$2,999 ($130,000-$155,999)' '$5,000-$5,999 ($260,000-$311,999)'
'$3,000-$3,499 ($156,000-$181,999)' '$6,000-$7,999 ($312,000-$415,999)'
'$300-$399 ($15,600-$20,799)' '$650-$799 ($33,800-$41,599)'
'$800-$999 ($41,600-$51,999)' '$4,500-$4,999 ($234,000-$259,999)'
'$1,000-$1,249 ($52,000-$64,999)' '$400-$499 ($20,800-$25,999)'
'$8,000 or more ($416,000 or more)' '$150-$299 ($7,800-$15,599)']

Total unique categories: 20

Unique values for Age Group:
['30->34' '25->29' '70->74' '55->59' '45->49' '15->19' '10->14' '60->64'
'65->69' '40->44' '75->79' '35->39' '20->24' '50->54' '5->9' '0->4'
'80->84' '85->89' '95->99' '90->94' '100+']

Total unique categories: 21
```

Figure 4: Unique values of household income groups and age groups

5. **Travel Mode Group:** An inspection of the travel mode variable identified a very small subset of records; only four entries were classified as ‘Other’. Given the extremely small sample size, this category was considered statistically insignificant for subsequent analysis and modelling. These records were therefore removed from the dataset to prevent skewing the results and to maintain a focus on the main travel mode groups (Private, Public, and Active). Including such a small category would have offered little analytical value and could have introduced unnecessary noise into model training.

Correlation

1. **Car ownership and Travel Mode:** Car ownership exhibited the highest NMI among all tested features with a score of 0.227. It implies that car ownership is the most influential factor affecting individual’s choice on travel mode. Household owning a car with at least one licenced member are much likely to travel in their private vehicle. The strong correlation also indicated its importance in the performance of machine learning models.

2. **Age and Travel Mode:** The NMI between age and travel mode was 0.06, indicating a relatively weak but notable correlation. It suggests that the age may have some influence on individual's choice of travel mode. According to assignment 1 task 4 (Figure 5), most journeys to education involved shorter commuting time, which increases the chance of walking rather than using private transportation.

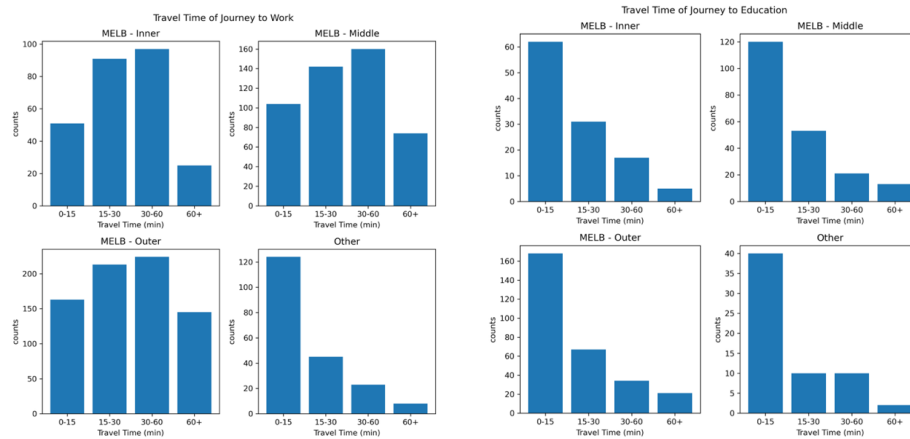


Figure 5: Bar charts illustrating travel time of residents in different regions on work journeys (left) and education journeys (right).

3. **Region and Travel Mode:** The NMI between region and travel mode was 0.04, indicating a weak but noteworthy relationship. The correlation between region and travel mode is likely related to the public transportation service provided in different regions. For example, in suburban areas where public transportation services are not sufficient, individuals would choose to drive to work to increase time efficiency and convenience.
4. **Employment Status and Travel Mode:** From employment status onwards, the NMI values have dropped further to 0.0277, showing a low correlation with travel mode. It may be due to the nature of study, as estimating behaviour of people from objective features often introduces noise into the data. Nevertheless, employment status remains its relevance to be regarded in the machine learning models. Intuitively, individuals who are unemployed or working part-time are less likely to use private travel mode due to its incurred cost.
5. **Travel Distance and Travel Mode:** The NMI between travel distance and travel mode was 0.0245, indicating a very weak correlation. Although this suggests that distance alone has limited ability to predict individual's travel mode choice, it remains a meaningful factor to explore as distance and region are the only two features which reflect geographical characteristics. Geographical factors are influential especially for population with long commuting range. This is evident from the working population, where the maximum travel distance exceeds 250km.
6. **Household Income and Travel Mode:** The NMI between household income level and travel mode was 0.0171, indicating an insignificant correlation between the two variables. From a real-world perspective, however, household income can be a dominant factor that determines the commuting options available to a household. It indirectly determines whether a household can afford a private vehicle, which affects the likelihood of selecting 'Private' mode. Simultaneously, it affects the choice of residential region which in turn affects travel distance.
7. **Bicycle Ownership and Travel Mode:** Households with bicycles may have a higher chance of choosing active modes to travel. However, the NMI value of 0.00484 indicated an extremely weak correlation. This suggests that even for households that own bicycles, they seldom choose them to be their primary mode of travel. Alternatively, the weak relationship might reflect indirect effects, where its influence is already captured by other features, such as travel distance. Overall, bicycle ownership appears to have weak contributing power on the model performance.

8. **Size and Travel Mode:** Similar to the previous factor, the NMI of household size was almost negligible, with a value of 0.0047. It indicates that household size has little or no correlation with travel mode. It may genuinely have no relationship with the outcome, or its effect may already be captured by other factors. As a result, household size is expected to have low predictive power and may be excluded from the model training.

Supervised Machine Learning

1. K-Nearest Neighbours (KNN)

The model was initially trained with $k = 5$ using the eight encoded features. On the hold-out set, the model achieved an accuracy of 0.753. Precision, recall, and F1 were 0.68, 0.753 and 0.706 respectively, which indicated a stable classification result despite the relatively simple hyperparameter choice. Thereafter, the 5-fold cross validation was conducted to tune the number of nearest neighbours in the model. By comparing the average accuracies across various values of k in each fold, the optimal value $k = 9$ (Figure 6 right) was identified. The KNN model was then refitted with the optimal value derived, yielding a slightly increased accuracy 0.759 on the testing set. Precision, recall and F1 score were 0.667, 0.759 and 0.698 respectively, demonstrating a stable classification after hyperparameter tuning.

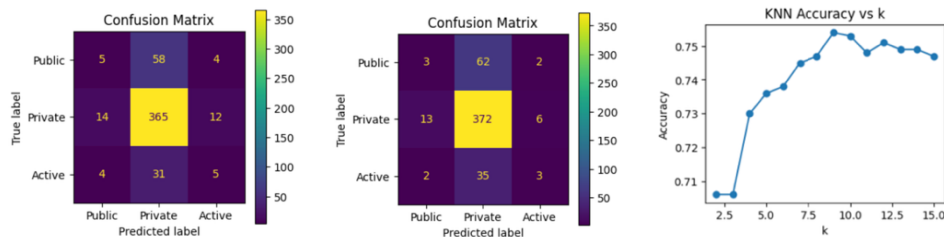


Figure 6: Confusion matrix before tuning k (left) and after tuning k (middle), and trend of average accuracies with different values of k during 5-fold validation (right)

As illustrated in the confusion matrices (Figure 6 left and middle), the dataset was skewed with 75% of outcomes fell into the 'Private' category. Thus, stratified sampling was adopted. Under this approach, the model accuracy decreased to 0.731, with precision, recall and F1 score of 0.651, 0.731 and 0.675 respectively and confusion matrix was reported in figure 7 left. Following this, the hyperparameter k was re-tuned. The optimal value shifted from 9 to 16 (Figure 7 right), and the selected optimal value was employed for model refitting. The performance of the KNN model remained stable with the same accuracy, precision, recall, F1 score, and confusion matrix (Figure 7 middle).

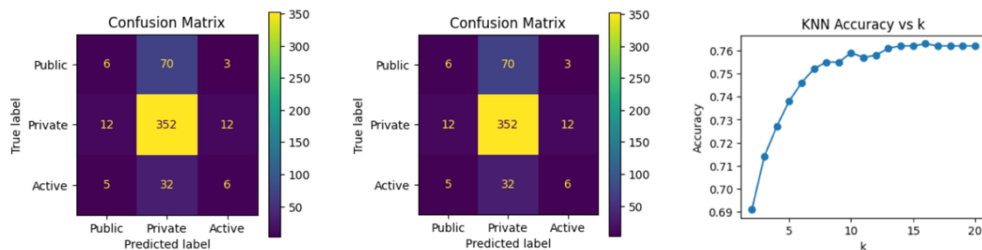


Figure 7: Confusion matrix with stratified sampling before tuning k (left) and after tuning k (middle), and trend of average accuracies with stratified sampling of different values of k during 5-fold validation (right)

From correlation analysis, some features were found to be insignificant in influencing the travel mode. Consequently, new models were trained after removing household size and bicycle ownership from the feature set. Following the previous procedures, the KNN model with $k = 5$ was first fitted and it yielded an accuracy of 0.733, with precision, recall and F1 score of 0.677, 0.733 and 0.695 respectively, and confusion matrix was reported in figure 8 left. Thereafter, the value of k was tuned using cross validation, and the optimal value identified was $k = 30$ (Figure 8 right). The model was then re-trained, and its performance

was evaluated. All four metrics remained the same compared to the initial hyperparameter together with the confusion matrix (Figure 8 middle). It shows that the model could capture the ‘Private’ mode accurately, but tended to over-predict it, leading to lower precision.

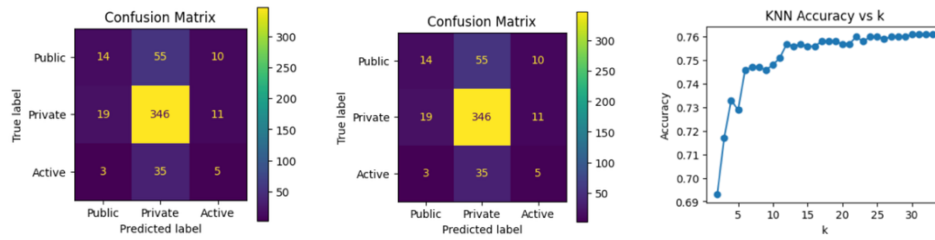


Figure 8: Confusion matrix with stratified sampling and feature selection before tuning k (left) and after tuning k (middle), and trend of average accuracies with further feature selection of different values of k during 5-fold validation (right)

2. Decision Tree

Similar to the KNN model, a decision tree model with the hyperparameter `max_depth = 3` was first trained on the hold-out set, which means the tree could make up to three levels of splits from the root node. When applied to test set, an accuracy of 0.801 was achieved with precision, recall and F1 score of 0.721, 0.801 and 0.727 respectively, and confusion matrix was reported in figure 9 left. From these evaluation metrics, the decision tree model performed better than the KNN model. However, the prediction did not include any ‘Active’ result. To optimize the model, different depth values (2, 3, 4, 5 and no limits) were tested using 5-fold cross validation. The results showed that the tree with `max_depth = 3` achieved the highest average accuracy (Figure 9 right), confirming that this setting can provide the best performance for the current dataset. The performance metrics remained stable (Figure 9 middle).

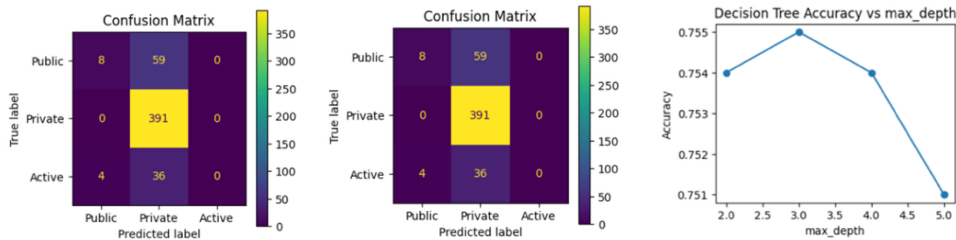


Figure 9: Decision Tree Model’s Confusion matrix before tuning `max_depth` (left) and after tuning `max_depth` (middle), and trend of average accuracies with different values of `max_depth` during 5-fold validation (right)

Since the randomly selected train dataset failed to predict ‘Active’ mode, stratified sampling strategy was employed. The model achieved a similar overall accuracy of 0.761, but the classification results became slightly more balanced. As shown in the confusion matrix (Figure 10 left), a small number of ‘Active’ samples were correctly identified. The tuning process was performed again by testing the `max_depth` values from 2, 3, 4, 5 to None, which confirmed that the model performed best when the depth was limited to 4 (Figure 10 right). The accuracy, precision, recall, F1 score were reported to be 0.759, 0.705, 0.759, 0.666, and confusion matrix was shown in figure 10 middle.

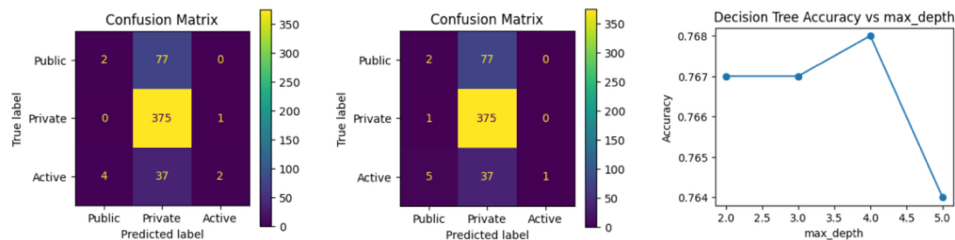


Figure 10: Confusion matrix with stratified sampling before tuning `max_depth` (left) and after tuning `max_depth` (middle), and trend of average accuracies with stratified sampling of different values of `max_depth` during 5-fold validation (right)

Based on the correlation analysis, the decision tree model was also retrained after removing features with lower correlation to the target variable. With $\text{max_depth} = 3$, the accuracy, precision, recall, F1 score were 0.771, 0.681, 0.771, 0.69 respectively and confusion matrix was reported in figure 11 left. Optimal value $\text{max_depth} = 3$ was identified in the tuning process (Figure 11 right). After feature selection, the overall accuracy remained similar to stratified sampling set with the same confusion matrix (Figure 11 middle). However, the model again failed to predict ‘Active’ mode.

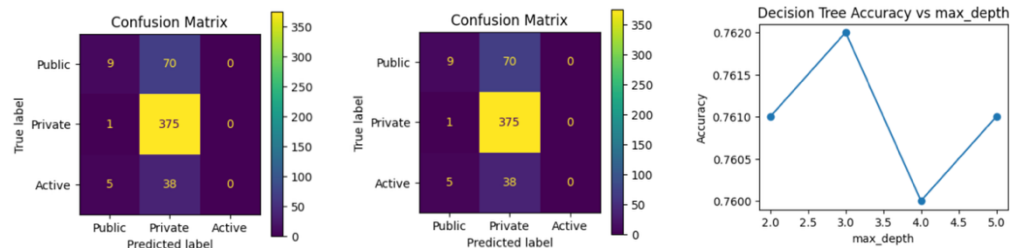


Figure 11: Confusion matrix with further feature selection before tuning max_depth (left) and after tuning max_depth (middle), and trend of average accuracies with stratified sampling of different values of max_depth during 5-fold validation (right)

Discussion and Interpretation

Data Preprocessing Insights

1. Missing Income Values Reflected True No-Income Households

During the stage of data exploration and data preprocessing, diagnostic merging between household and person datasets revealed that most missing income records belonged to households where no member reported personal income or employment. This indicated systematic missingness, not random data loss. Treating them as zero-income rather than imputing by mean or median preserved the socio-economic meaning and avoided bias. This finding highlights how thoughtful preprocessing can prevent misinterpretation of low-income households as missing data. It also confirms that employment and income are strongly linked structural determinants of travel access.

2. Outlier Assessment and Data Integrity

Outliers such as negative journey distances were removed, while unusually large household sizes were retained to capture multi-generational and extended family structures, which is common in some Melbourne regions. These preprocessing choices were crucial to ensure that downstream analyses reflected realistic population diversity, particularly across regional and income lines. It established that socioeconomic and diversity in the dataset is not noise but a crucial feature of Victoria’s travel behaviour landscape.

Correlation Insights

1. Car Ownership as Dominant Predictor

Car ownership showed the highest NMI of 0.227 with travel mode. Individuals in car-owning households overwhelmingly preferred private travel modes, while those without access were constrained to public or active transport. This finding reflects the structural car dependency long documented in Melbourne. This is further supported by Dovey et al. (2015), they found that car users experienced four to sixteen times greater reachability than public transport users, illustrating how network design reinforces private vehicle dominance. Thus, car ownership in the models captures not only personal preference but also reflects systemic inequalities in accessibility embedded in Melbourne’s urban form.

2. Age and Life-Stage Effects

Age demonstrated a moderate correlation (NMI = 0.06), where younger individuals and students tended to favour public or active modes, whereas working-age adults predominantly used private transport. This

pattern mirrors life-stage differences in mobility demand and resource access, where younger individuals often lack vehicle ownership but have shorter or education-related trips.

3. Regional Factors

Although regional location showed only a weak direct correlation ($NMI \approx 0.04$), it clearly shapes how people travel. Inner Melbourne residents have more access to trains, trams, and walkable areas, while those in outer or regional suburbs rely heavily on cars due to limited public transport options. According to Gharehbaghi et al. (2020), transport infrastructure affects accessibility, economic opportunity, and quality of life. When public transport is infrequent or poorly connected, people lose access to jobs, education, and services, leaving private vehicles as the only practical option. In contrast, well-connected areas allow more flexible and sustainable travel choices. This also further explains the strong importance of car ownership as it reflects the structural differences in transport access between Melbourne's inner and outer regions.

4. Income as Indirect Drivers

Employment status exhibited a weak but non-trivial relationship with travel mode ($NMI = 0.0277$). Part-time or unemployed individuals were more likely to use public or active modes, potentially due to cost sensitivity or lack of car access. Income, however, displayed an unexpectedly low direct correlation ($NMI = 0.0171$). Indeed, Nguyen et al. (2017) also proved that income's effect on mode choice is largely indirect, operating through mediating factors such as residential location and vehicle ownership. Moreover, they observed a nonlinear mediated effect - as middle-income groups are more likely to live in areas with better public transport access and therefore use it more often, while higher-income households tend to reside in car-dependent suburbs, leading to increased private vehicle use. Therefore, our weak income correlation is consistent with this interpretation income does not independently drive travel mode choice but operates through its effects on housing location and car ownership, producing an indirect and non-linear relationship with mode choice.

5. Household Size and Bicycle Ownership Insignificance

Both household size and bicycle ownership showed near-zero correlation with travel mode ($NMI < 0.005$), suggesting that neither factor has a strong influence on how individuals choose their mode of travel. This outcome may seem surprising, but it reflects the way broader social and environmental factors shape travel behaviour. As Zhao et al., (2023) explained, population structure, including household composition, income, and regional demographics, affects transport demand by influencing accessibility and the distribution of infrastructure investments, rather than influencing individual mode choices directly. Larger households may own more vehicles or include members with differing mobility needs, but these effects are mediated through car ownership and residential location, not only household size.

Similarly, Olayode, et al., (2024) found that in Melbourne's western suburbs where bicycle ownership is common, low levels of walking and cycling persist due to poor network connectivity, safety concerns, and limited infrastructure support. Their findings demonstrate that owning a bicycle does not necessarily translate to using it for commuting, especially when local environments are car-oriented and lack of safe, continuous cycling routes. Altogether, these results suggest that both household size and bicycle ownership are context-dependent factors. Their influence on travel behaviour is largely determined by external conditions, such as urban design, transport accessibility, and socio-economic context, rather than by the characteristics themselves.

Supervised Machine Learning Interpretation

1. KNN Model Discussion

First, since the dataset was skewed towards the 'Private' class, stratified sampling was employed. It reduced the accuracy from 0.759 to 0.731. Although the drop of accuracy might seem detrimental, the risk of overfitting was prevented as the class distribution was maintained during the training and testing process.

In the initial split, the model could achieve high accuracy simply by predicting the majority class, which would apparently be biased. Despite the decline in the performance metrics, stratified sampling still provides a fairer view of actual travel behaviour.

Second, feature selection was performed using the correlation analysis. After removing household size and bicycle ownership, the accuracy was 0.733, similar to the previous results. It indicated that the two features had little contribution to the performance of model, supporting the removal of the variables. The reason is that the two variables have low predictive power compared to the other two features. A simpler model was therefore preferable as it maintained accuracy while reducing model complexity.

Hyperparameter tuning was done in each training stage to optimise performance. However, while $k = 30$ yielded the highest average accuracy during cross validation, it might be excessively too large that led to over-smoothing and reduced the generalising power of the model. The unchanged performance metrics indicated that fewer neighbours was sufficient to provide accurate classification.

2. Decision Tree Model Discussion

The decision tree model demonstrated relatively strong predictive performance compared with KNN model, achieving higher overall accuracy and balanced precision–recall trade-offs. However, across different experiments, the model consistently struggled to predict the ‘Active’ travel mode, indicating a sensitivity to data imbalance. When stratified sampling was applied, the classification results became more balanced, confirming that proportional representation of travel modes can help mitigate bias and enhance ability to capture infrequent categories like ‘Active’ mode. Nevertheless, the slight decrease in evaluation metrics suggests a trade-off between fairness and predictive power.

Feature selection analysis revealed that even predictors with low mutual information provided subtle but meaningful information related to less common travel modes. When these variables were removed, the model relied more heavily on dominant predictors such as car ownership, thereby increasing bias towards majority classes. These findings highlight both the interpretability and the limitation of decision tree models when applied to imbalanced categorical data.

3. Supervised Machine Learning Models Comparison

Regarding overall predictive performance with their respective optimal hyperparameters, the decision tree model achieved higher accuracy, precision, recall and F1 score than KNN model. This suggests that decision tree model can deliver more precise and stable predictions than KNN model, regardless of whether the dataset was randomly sampled or stratified, even when fewer predictive features were included.

However, when examining the confusion matrices, KNN model was more effective in capturing the minority ‘Active’ mode. In contrast, the decision tree model failed to identify ‘Active’ data samples and showed a strong bias toward the dominant ‘Private’ group especially when using randomly selected samples. Even though the decision tree model successfully predicted ‘Active’ mode when using stratified sampling, its accuracy was still insufficient compared with KNN model and other mode groups, making it less effective in representing the model’s prediction fairness. This finding indicates that KNN model demonstrates better robustness in handling imbalanced data distributions, as its distance-based approach allows it to recognize smaller and more dispersed patterns that are often overlooked by tree-based model.

Overall, the results indicate that KNN model has better adaptability to imbalanced dataset, whereas the decision tree model performs better in terms of overall prediction accuracy.

Limitations and Improvement Opportunities

From the data perspective, the features selected are mainly related to distance and finance related factors, but intuitively an individual’s travel mode also can be influenced by various social and behavioural factors, such as lifestyle preferences, environmental awareness etc which are not shown in the original dataset. The absence of these dimensions may limit the model’s performance on fully explaining and predicting travel

behaviour. In addition, the obvious imbalanced data sample distribution and high dependence on car ownership causes the current model prediction bias toward 'Private' mode.

From the methodological perspective, the major limitation of KNN model is that it is sensitive to feature scales and data noise. Since KNN relies on distance between samples, thus features with large numerical ranges tend to dominate the predication if the data failed to be normalised properly. This issue and the computational cost will be magnified when applied in high-dimension feature. Data standardization and dimension reduction technique such as principal components analysis (PCA) can be applied to address this issue. For Decision Tree model, one obvious disadvantage is its tendency to overfit the training data and instability to small data deviation. It also fails to capture linear relationship or intersection between features, as the model performs classification by splitting the feature space. This characteristic may also lead to bias toward multi-valued features. Random Forest or Gradient Boosting Decision Tree models can be adopted to address the model's inherent limitations. Oversampling or undersampling can be applied to handle imbalanced datasets, and pre-pruning or post-pruning techniques can be used to prevent from overfitting.

Conclusion

This report examined the effects of various factors influencing individual choices of travel mode, supported by correlation analysis using normalised mutual information and supervised machine learning using k-nearest neighbours and decision tree models. Furthermore, stratified sampling and feature selection were employed during model training process.

Throughout the study, car ownership was found to be the most influential factor, while household size and bicycle ownership had the least importance in choosing travel mode. This indicates that individuals are more likely to travel with their private vehicle if they have a licenced member. The use of stratified sampling improved the representativeness of classification models when applied on an imbalanced dataset. The reduction of accuracy is expected as a trade-off of fairness and predictive power. Generally, decision tree model has better performance regarding the accuracy, precision, recall and F1 score. However, it shows a strong bias towards the dominant class. On the other hand, KNN model yields lower accuracy, but it is less sensitive to the skewness of the data and thereby produce a fairer result.

These results have contributed to have a better understanding of individual travel behaviour. Further study is needed to explore the factors affecting travel mode, especially public and active, so that future policies can be established to improve travel efficiency and support environmental protection.

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